Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/28/2007
PI Name:	Glotzer, Sharon C Ph.D.		
Project Title:	ROLE OF SPATIALLY HETEROGENEOUS DYNAMICS AND POLYDISPERSITY IN THE EARLY STAGES OF HOMOGENEOUS NUCLEATION: APPLICATION TO THE PROMOTION AND SUPPRESSION OF CRYSTALLIZATION AND VITRIFICATION		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	48109-2136	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	3
No. of PhD Candidates:	5	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Solomon, Michael (University of Michigan)		
Grant/Contract No.:	NNC04GA43G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This task carries out ground-breaking research on the solidification behavior of supercooled liquids and colloids above their glass transition using molecular dynamics computer simulations and confocal laser scanning microscopy experiments. Homogeneous nucleation phenomena and its relation to local structural and dynamical features is studied. Nucleation and growth and the formation of quasicrystals is explored. This project addresses fundamental, ground-breaking issues in the solidification of liquids, particularly bulk metallic glass-forming liquids, and colloidal suspensions. The results of this project are critical to understanding and controlling the properties and behavior of technologically-relevant materials, in particular for space applications. Our work will provide strategies for designing colloidal materials with specific interactions to cause their solidifcation via self-assembly into unique crystal structures for target applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	New materials designed for space exploration and long-term space travel have strategic requirements related to strength, weight, durability, cost, etc. The science of solidification provides the necessary understanding for controlling the properties and behavior of technologically-important materials and making new materials for space exploration.
Task Progress:	1. Science Progress and Status In this project, we seek to elucidate the initial stages of crystal nucleation in supercooled liquids and colloids above their glass transition, and to relate nucleation phenomena to local structural and dynamical features in these systems. We further seek to understand the role of discrete size polydispersity in suppressing nucleation, and to compare this phenomenon in colloids and a model bulk metallic glass-forming liquid that forms quasicrystals as a metastable phase. Our goals in the initial year of the grant focused on carrying out initial simulations of a model supercooled liquid to characterize the spatially heterogeneous nature of the dynamics and relate it to local structure, and develop and test the analysis codes needed to quantify local order for studies of nucleation. Experimentally, our goals included establishing the experimental tools and parameters for carrying out confocal laser scanning microscopy (CLSM) experiments and conducting initial shear experiments. In Year 2 of the grant, we focused on developing a model pair potential for a monodisperse colloidal system that can be synthesized as well as simulated. As a first step, we developed a plan to create an experimental system that models the potential that we are also simulating. The colloidal equivalent potential is being parameterized in terms of the potential to develop a method for quantitative comparison between theory and simulation. Once the parameterization is in hand, colloids will be synthesized and characterized via CLSM. We used the local structural and dynamical computational analysis tools developed in Y1 to analyze local structure relevant for nucleation and crystallization, including vacancies, stacking faults, dislocations and grain boundaries in crystals formed via shear and sedimentation. In the final year, Year 3, of the grant, we showed that quasicrystal growth occurs via a novel mechanism whereby locally ordered groups of atoms in low energy configurations (here, icosahedral) are assimilated into the growing quasicrystal nucleus without rearrangement, thereby producing an aperiodic crystal. We also showed the strings of highly mobile particles in supercooled liquids are correlated to local structural signatures in the liquid. We investigated shear-induced colloidal crystallization of close-packed structures and reported a novel measure of close-packed layer structure heterogeneity in colloidal crystals by analogy with a standard copolymer measure of sequence randomness. We analyzed data from experiments on an air-driven granular system and showed spatially heterogeneous dynamics that are virtually indistinguishable from that in simulated supercooled liquids. Finally, we carried out a joint simulational-experimental investigation of pair correlation functions in dilute colloidal suspensions and investigated the role of polydispersity.
Bibliography Type:	Description: (Last Updated: 04/09/2008)
Articles in Peer-reviewed Journals	Gebremichael Y, Vogel M, Bergroth MN, Starr FW, Glotzer SC. "Spatially heterogeneous dynamics and the Adam-Gibbs relation in the Dzugutov liquid." J Phys Chem B. 2005 Aug 11;109(31):15068-79. PMID: 16852907 , Aug-2005
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Articles in Peer-reviewed Journals	Keys AS, Abate A, Glotzer SC, Durian DJ. "Measurement of growing dynamical length scales and prediction of the jamming transition in a granular material." Nature Physics, 2007 Apr;3(4):260-4. http://dx.doi.org/10.1038/nphys572 , Apr-2007
Articles in Peer-reviewed Journals	Glotzer SC, Solomon MJ, Kotov NA. "Self assembly: From nanoscale to micron scale colloids." AIChE Journal 2004 Dec;50(12):2978-85. http://dx.doi.org/10.1002/aic.10413 , Dec-2004
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